Vaishnavi Hanamant shinde

bm25mtech11009@iith.ac.in | shindevaishnavi781@gmail.com | +91 8830508044 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.04
Mahatma Gandhi uchha madhyamik vidyalaya, Umbraj	Class XII (Maharashtra state board of secondary and higher secondary education, pune)	2021	79.33%
Rukminibai pandurang kadam kanya vidyalaya, Umbraj	Class X (Maharashtra state board of secondary and higher secondary education, pune)	2019	84.80%

PROJECTS

Medical Insurance Cost Prediction

- Developed an end-to-end **Python ML regression pipeline** using Linear Regression, Random Forest, and **XGBoost**, achieving **0.954 R^2 and $1,591 MAE**.
- Implemented **Scikit-Learn Gradient Boosting quantile models** to deliver calibrated **90% prediction intervals with 87.7% empirical coverage**.
- Used **SQL analytics queries** and **SHAP TreeExplainer** to quantify key cost drivers, and deployed an interactive **Streamlit** application.

Nanoparticle Toxicity Risk Prediction

- Performed **data preprocessing and feature analysis using Python and Pandas** across 900 nanoparticle formulations with 8 physicochemical features.
- Built **Scikit-Learn classification models** using Logistic Regression and Random Forest, achieving **95.6% accuracy, 0.966 ROC-AUC, and 98.2% recall**.
- Conducted **statistical analysis using Chi-Square testing** and Gini feature importance to identify toxicity risk drivers and assess the protective effect of **PEGylation coatings**.

Student Exam Performance Prediction

- Built a **Python regression pipeline** predicting student academic scores with over **85% R^2 across 1,000 records**.
- Implemented automated data preprocessing using **Scikit-Learn ColumnTransformer** for missing-value imputation and **one-hot encoding**.
- Evaluated **7+ regression models** using **GridSearchCV and XGBoost**, optimizing model performance through hyperparameter tuning.

Pharmacy Demand & Inventory Analytics | Python, Scikit-Learn, SQL, Power BI, Pandas

- Cleaned **29,000+ transaction rows using Pandas & NumPy** by fixing nulls, duplicates, and format anomalies.
- Executed **ABC analysis & inventory risk queries via SQL**, isolating 17 Class-A SKUs driving **69% revenue**.
- Built **Scikit-Learn regression models with Fourier terms**, reducing demand forecast error by **29%-57% MAE**.

SKILLS

Programming: Python, R Programming, SQL

Machine Learning & Analytics: Machine Learning, Statistics, Predictive Modeling

Data & Visualization: Pandas, NumPy, Matplotlib, Power BI, Tableau

Productivity & AI: Excel with AI, Prompt Engineering

Tools & Platforms: GitHub, MySQL, PostgreSQL, Jupyter Notebook, Google Colab

Management: Team Management

RELEVANT COURSES

Machine Learning and its application - by **Dr. Venkatesham B** (Mae dept)and **Dr. R Prasanth Kumar** (AI dept)

Data Science (Tutedude): Completed practical training in Python, data analysis, machine learning, and data visualization, applying these skills to real-world datasets and analytical problems.

POSITIONS OF RESPONSIBILITY

PG Mentor -- **Sunshine**

- Guided PG students in academic and personal development through regular one-on-one support and mentorship.

- Addressed student concerns and fostered a supportive, positive, and collaborative living environment.

CERTIFICATIONS

Deloitte Data Analytics Job Simulation -- **Forage**

Advanced SQL -- **Tutedude**

Google Data Analytics -- **Coursera**

EXTRACURRICULAR

Cultural Activities:

- Participated in traditional dance performances during **Sankranti and Onam celebrations**.

HOBBIES

- Dancing, Drawing ,Cooking